This Is the Moment

Li-Yuan Chiang
with David Poland and Gordon Rogelberg

Yale University

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Based on *Moments in the CFT Landscape*, arXiv:2603.18140

In a nutshell...

- Crossing symmetry and unitarity constrain not only individual operators, but also weighted statistics of the OPE.
- Moment variables play a central role, which remain effective for heavy correlators (large external scaling dimensions) and light correlators.
- We discovered a beautiful, less-explored side of the conformal bootstrap landscape shaped by spectral reorganization.

Implications of conformal symmetry

- Local operators are organized into conformal multiplets. Primary operators are labeled by their **scaling dimension** and **spin**:

$$\mathcal{O}_i \quad \longleftrightarrow \quad (\Delta_i, \ell_i).$$

- Conformal symmetry fixes correlation functions up to 3-pt:

$$\langle \phi(x_1)\phi(x_2)\rangle = \frac{1}{x_{12}^{2\Delta_\phi}}, \quad \langle \phi(x_1)\phi(x_2)\phi(x_3)\rangle = \frac{\lambda_{123}}{x_{12}^{2\alpha_{123}} x_{23}^{2\alpha_{231}} x_{31}^{2\alpha_{312}}},$$

- The convergent **Operator Product Expansion** (OPE)

$$\phi(x)\phi(y) = \sum_{\mathcal{O}\in\phi\times\phi} C_{\mathcal{O}}(x-y, \partial_y)\mathcal{O}(y)$$

reduces higher-point functions to the **conformal data** $\{(\Delta_i, \ell_i), \lambda_{ijk}\}$.

Four-point functions and conformal blocks

We will focus on 4-pt functions of identical scalar primaries:

$$\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle = \frac{1}{x_{12}^{2\Delta_\phi} x_{34}^{2\Delta_\phi}} G(u,v),$$

where

$$u = z\bar{z} = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \qquad v = (1-z)(1-\bar{z}) = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}.$$

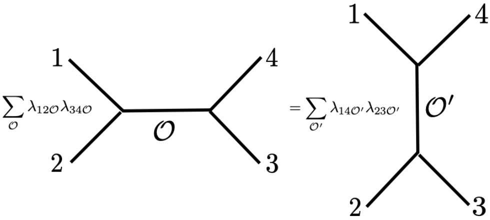
The OPE gives a partial-wave expansion:

$$G(u,v) = \sum_{\mathcal{O}\in\phi\times\phi} \lambda_{\phi\phi\mathcal{O}}^2 g_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v).$$

The **conformal blocks** package the contribution of a primary and all of its descendants: they are the "spherical harmonics" of a CFT.

Crossing symmetry

This is not the only way we can perform the OPE!



Different OPE decompositions of the same correlator must agree. For identical scalars,

$$0 = \sum_{\mathcal{O} \in \phi \times \phi} \lambda^2_{\phi\phi\mathcal{O}} F_{\Delta_{\mathcal{O}}, \ell_{\mathcal{O}}}(z, \bar{z}),$$

$$F_{\Delta,\ell}(z, \bar{z}) = u^{-\Delta_\phi} g_{\Delta,\ell}(u, v) - v^{-\Delta_\phi} g_{\Delta,\ell}(v, u).$$

The conformal bootstrap

- The **Bootstrap Equation**:

$$0 = \sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}} \left[u^{-\Delta_\phi} g_{\Delta,\ell}(u,v) - v^{-\Delta_\phi} g_{\Delta,\ell}(v,u) \right],$$

- Constraints from **unitarity**:

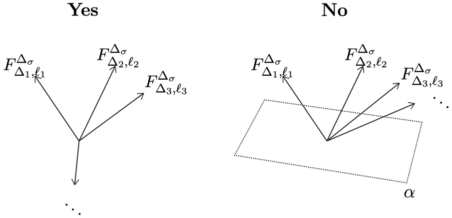
$$\Delta \geq \ell + \frac{D-2}{2}\left(2 - \delta_{\ell,0}\right).$$

- Infinite number of unknowns ($\{\Delta_{\mathcal{O}}, \lambda_{ij\mathcal{O}}\}$), infinite number of constraints ($\forall z, \bar{z}$)!
- How far can we go with these *consistency conditions* alone?

Extremal functional method

$$0 = \sum_{\mathcal{O}\in\phi\times\phi} \lambda^2_{\phi\phi\mathcal{O}} F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z}),$$

1. Make a guess for the conformal data (e.g., assuming a gap).
2. Look for a linear combination of $\partial_z^m \partial_{\bar{z}}^n F_{\Delta,\ell}$ such that it is *positive*.
3. If such a solution exists, the proposed data is **incompatible** with crossing and unitarity!

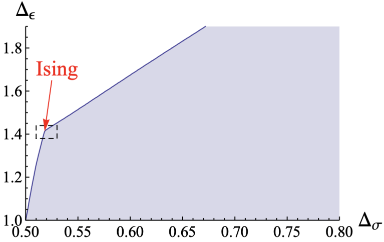


The Conformal Bootstrap, DOI: 10.1038/NPHYS3761

In practice, this can be solved with semidefinite programming (SDP).

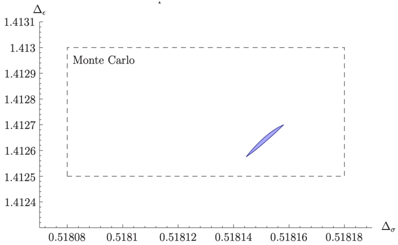
Carving out the theory space

Single-correlator bootstrap
Upper bounds on the leading scalar dimension have a kink remarkably close to the 3D Ising!



arXiv:1203.6064

Mixed-correlator bootstrap
Additional crossing constraints + 2 relevant operators



arXiv:1502.02033

Theory space is usually described using properties of the low-lying operators: $\Delta_{\text{gap}}, \lambda_{\phi\phi\mathcal{O}}, c_T, \ldots$

New perspective: moment variables

- Low-lying observables work well when the spectrum is sparse.
- For dense/heavy spectra, the relevant OPE weight may be distributed over many operators. $\Rightarrow$ Tracking each operator is expensive (requires many derivatives).
- Becomes specially serious at large external dimension Δ_ϕ.

Our angle

Instead of asking only where the first operator lies, constrain **weighted statistics**, i.e. **moments**, of the entire exchanged spectrum.

Can the bootstrap directly constrain these average properties of the OPE distribution?

New perspective: moment variables

- We define moments as weighted averages over the conformal data, with the spectral density

$$p_\ell(\Delta) = \sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}}\, g_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z})\delta(\Delta - \Delta_{\mathcal{O}})\delta_{\ell,\ell_{\mathcal{O}}}$$

  This defines a positive measure when the theory is unitary and the conformal block is positive.

- We will focus on moments computed at the self-dual kinematics $z = \bar{z} = 1/2$ from now on, where the conformal block is positive.

New perspective: moment variables

- Moments are defined as the following weighted averages (normalized by the correlator):

$$\nu_{m,n} = \sum_{\ell} \int d\Delta p_{\ell}(\Delta) \Delta^m \ell^n = \sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}}\, g_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z}) \Delta^m_{\mathcal{O}} \ell^n_{\mathcal{O}}|_{z=\bar{z}=1/2}$$

$$\langle \Delta^m \ell^n \rangle \equiv \frac{\nu_{m,n}}{\nu_{0,0}}, \quad \langle \Delta^m \ell^n \rangle_* \equiv \frac{\nu_{m,n}}{\nu_{0,0} - 1}$$

- We will use the shorthand notation $\nu_m = \nu_{m,0}$.
- Now, we can compute the "average" scaling dimension of your favorite CFT and even describe how the operators "spread".
  $\rightarrow$ In statistics, they are given by $\langle \Delta \rangle$ and $\langle \Delta^2 \rangle - \langle \Delta \rangle^2$.

Why moments?

- They are global observables over the entire OPE
- They are the statistics of the primaries
- They are numerically friendly (SDP techniques available!)

Moments sacrifice microscopic resolution, but remain meaningful when the spectrum becomes dense.

Bounding moments

$$\text{Maximize} \quad \sum_{\ell} \int_{\Delta_{\min}(\ell)}^{\infty} d\Delta\, p_\ell(\Delta) V_\ell(\Delta)$$

$$\text{subject to} \quad \sum_{\ell} \int_{\Delta_{\min}(\ell)}^{\infty} d\Delta\, p_\ell(\Delta) F_\ell^{(\alpha)}(\Delta) = v^{(\alpha)}, \quad p_\ell(\Delta) \geq 0,$$

$$\text{Minimize} \quad \vec{v} \cdot \vec{z}$$

$$\text{subject to} \quad \vec{z} \cdot \vec{F}_\ell(\Delta) - V_\ell(\Delta) \geq 0 \quad \forall \Delta \geq \Delta_{\min}(\ell) \text{ and } \ell \in \mathcal{S},$$

$$\vec{v} \cdot \vec{z} = \vec{z} \cdot \sum_{\ell} \int_{\Delta_{\min}(\ell)}^{\infty} d\Delta p_\ell(\Delta) \vec{F}_\ell(\Delta) \geq \sum_{\ell} \int_{\Delta_{\min}(\ell)}^{\infty} d\Delta p_\ell(\Delta) V_\ell(\Delta)$$

Bounding moments

$$\begin{aligned} \text{Maximize} \quad & \sum_\ell \int_{\Delta_{\min}(\ell)}^\infty d\Delta\, p_\ell(\Delta) V_\ell(\Delta) \\ \text{subject to} \quad & \sum_\ell \int_{\Delta_{\min}(\ell)}^\infty d\Delta\, p_\ell(\Delta) F_\ell^{(\alpha)}(\Delta) = v^{(\alpha)}, \quad p_\ell(\Delta) \geq 0, \end{aligned}$$

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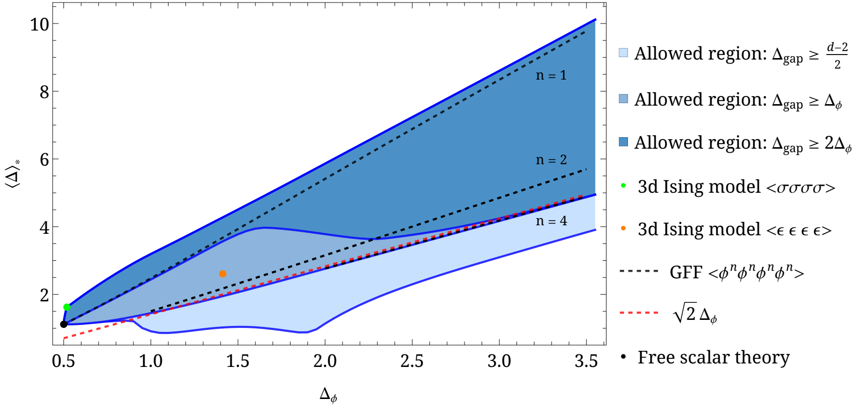
- Similar logic can be used to bound the normalized moments.
- By approaching the boundary of the allowed region, we can extract extremal spectra where the inequality is saturated.

Moment bounds in 3D



The heavy-correlator limit

- Asymptotic conformal blocks + crossing + unitarity

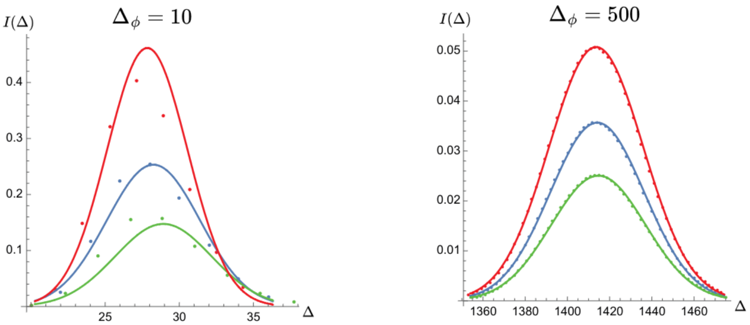
$$\Rightarrow (\sqrt{2}\Delta_\phi)^k + \mathcal{O}(\Delta_\phi^{k-1}) \leq \langle \Delta^k \rangle \leq \frac{1}{2}(2\sqrt{2}\Delta_\phi)^k + \mathcal{O}(\Delta_\phi^{k-1})$$

  in the large Δ_ϕ limit.
- The bounds are saturated by the generalized free field (GFF). GFF correlators can be computed by Wick contractions:

$$G_{GFF}(z,\bar{z}) = \frac{\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle}{\langle \phi(x_1)\phi(x_2)\rangle\langle \phi(x_3)\phi(x_4)\rangle} = 1 + u^{\Delta_\phi} + (u/v)^{\Delta_\phi}.$$

The Heavy-Correlator Limit

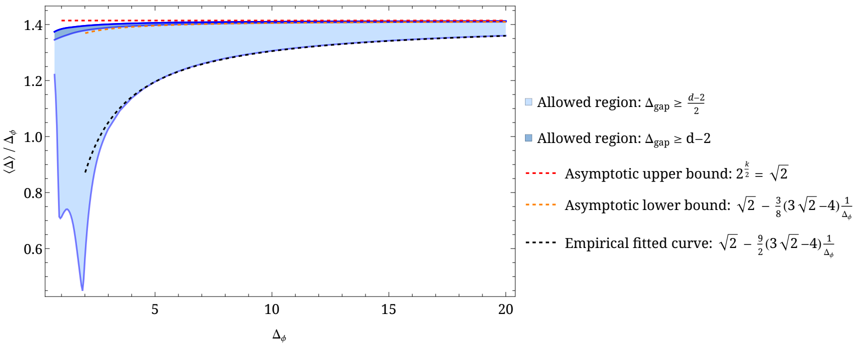
Poland and Rogelberg, arXiv:2501.00092



arXiv:2501.00092

- The dominant contribution no longer comes from a few isolated light states, but from the **collective behavior** of heavy operators.
- The heavy-correlator regime was long considered a challenging regime for the numerical bootstrap.

Recovering Analytical Results — And More!

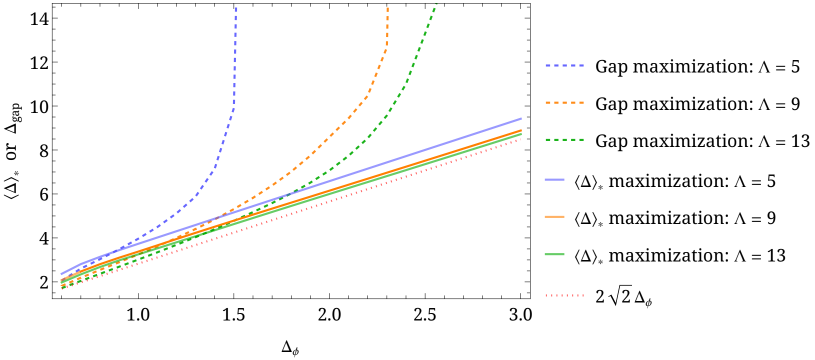


Numerical bounds approach the asymptotic bounds rapidly!

Highly nonlinear structures appear at small Δ_ϕ!

Moment maximization vs. Gap maximization

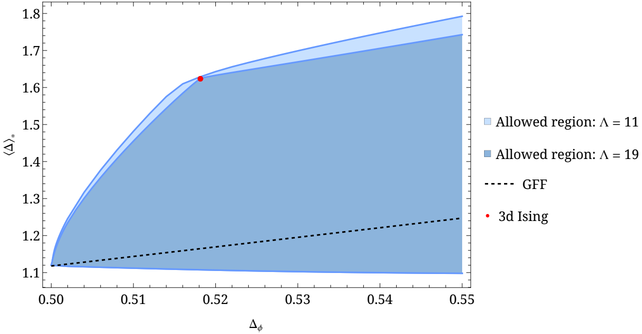
This is not a fair comparison, but the convergence rate demonstrates that moments are more natural observables in the heavy limit.



The numerical bootstrap "knows" how to use the derivative budget wisely, placing a few states accurately so that it reproduces the correct leading moment bounds.

3D Ising revisited

Maximizing $\langle \Delta \rangle_*$, we "rediscover" the famous 3D Ising kink.



The OPE is essentially dominated by the gap state: $\langle \Delta \rangle_* \approx \Delta_{\text{gap}}$.

Other moments also develop a kink at the same location.

Precise Ising moments

With the following scaling dimensions we obtained ($\Lambda = 27$)

$$\Delta_\sigma = 0.5181488, \quad \Delta_\epsilon = 1.412625,$$

Moment	Bounds	Known data ($\Lambda = 43$)
$\nu_{0,0}$	0.768530(25)	0.7685483(69)
$\nu_{1,0}$	1.247745(15)	1.247736(21)
$\nu_{0,1}$	0.2004013(17)	0.200381(14)
$\nu_{2,0}$	2.2725170(20)	2.272380(91)
$\nu_{1,1}$	0.6255126(41)	0.625408(68)
$\nu_{0,2}$	0.4246333(36)	0.424550(54)

The bounds are extremely tight and even exclude the values computed from the stable operators in arXiv:1612.08471.

$\Rightarrow$ This points toward omitted higher-twist or unstable operators!

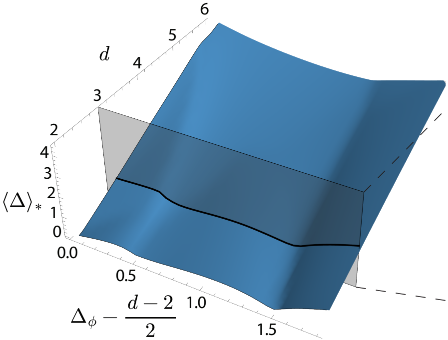
The other side of the bootstrap landscape...

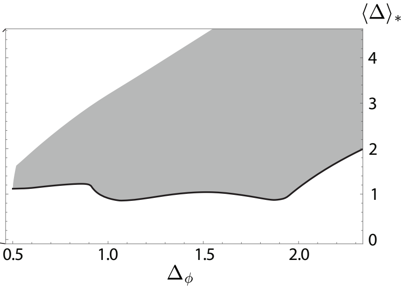
The results are encouraging, but the real surprise comes when we examine the **lower bounds** on the moments!

We discovered structures that, to the best of our knowledge, have not been observed in previous bootstrap literature.

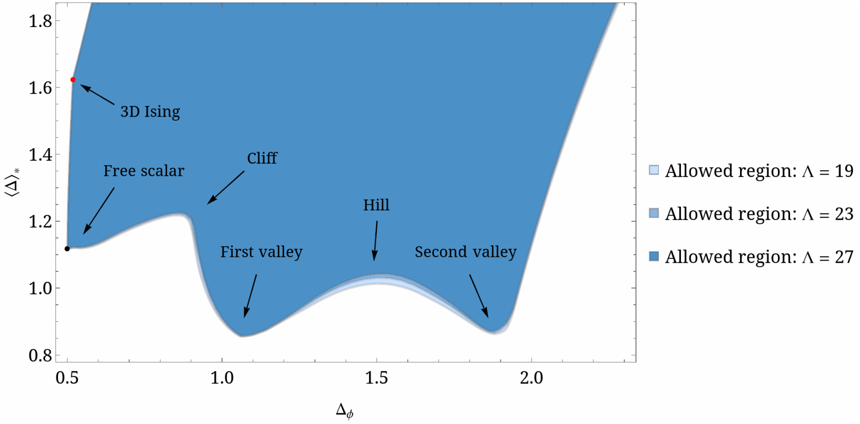
The physical origin remains unknown, but recent progress has improved our understanding of their spectral behavior.

A beautiful landscape





Welcome to the CFT Park!



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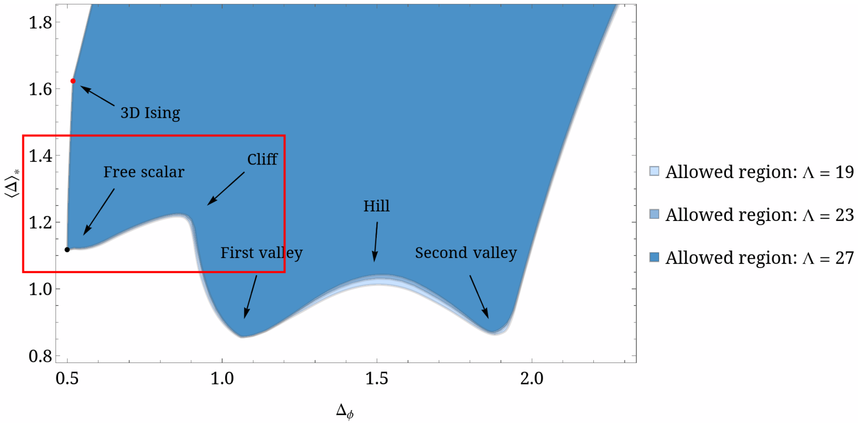
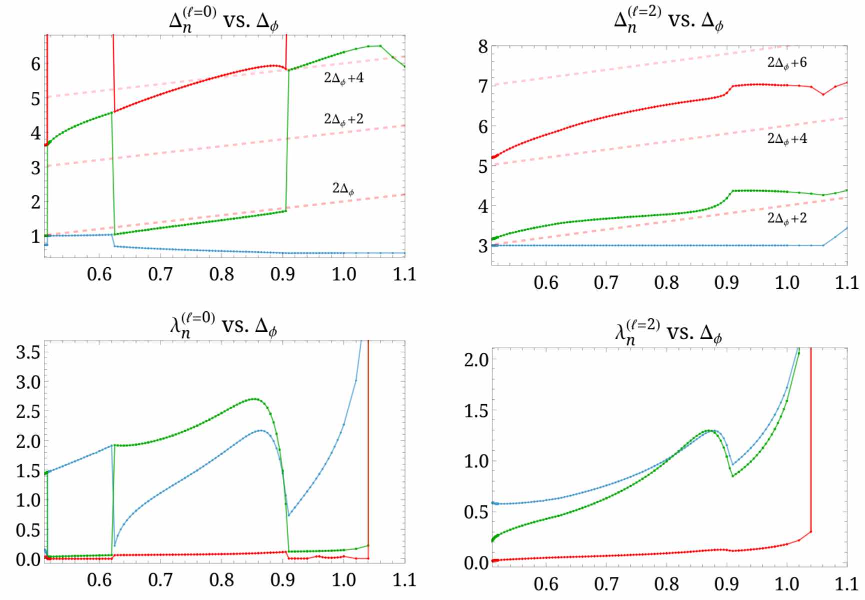


Extremal spectra before the "valley"



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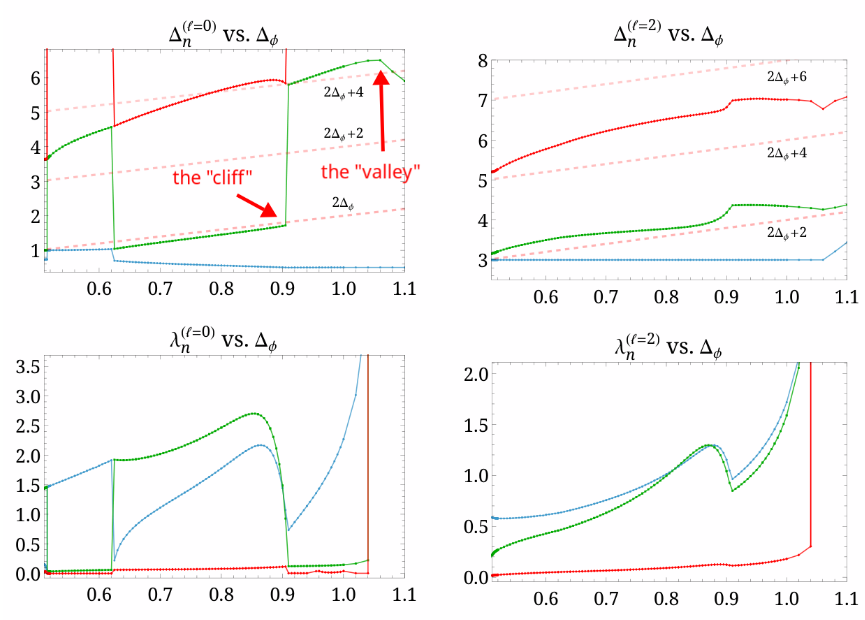
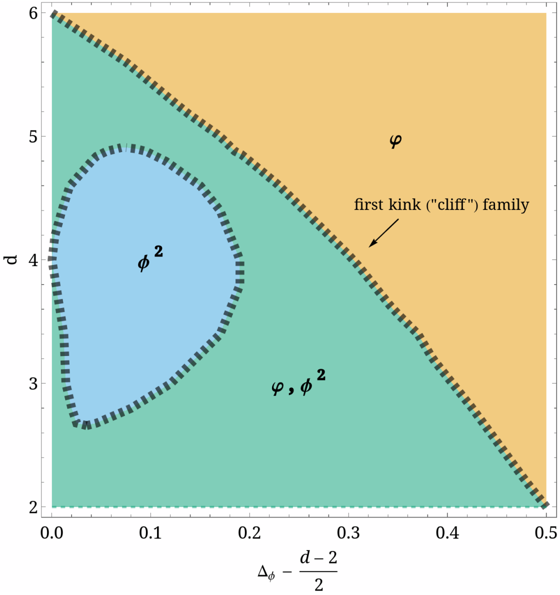




φ
first kink ("cliff") family
ϕ^2
φ, ϕ^2
d
$\Delta_\phi - \frac{d-2}{2}$

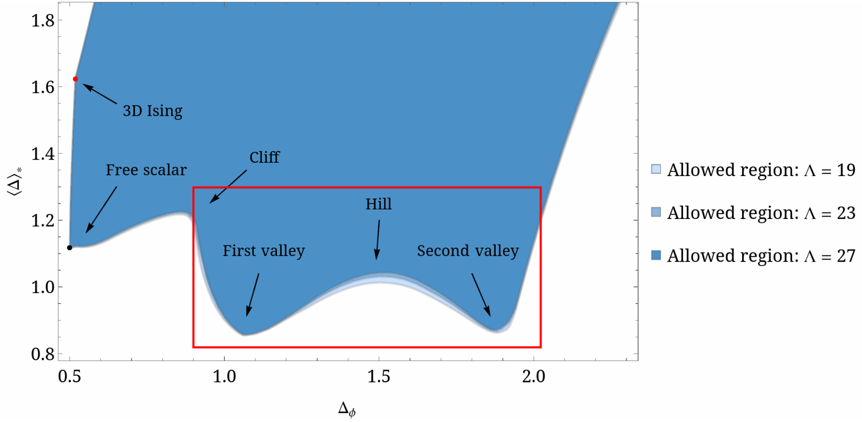
Open question

Kos et al, arXiv:1504.07997

- Kinks are tied to the disappearance of certain operators in the extremal spectra, i.e., **operator decoupling**, also observed in the Wilson-Fisher description of the Ising model.
- Operator decouples at $2\Delta_\phi$ + connected to 6D free scalar
  $\rightarrow$ Could this reflect an EOM of the form $\Box\phi \sim \phi^2$?

Do these structures arise from renormalization group flows of known interacting scalar theories, or instead probe an approximately unitary sector of more exotic solutions?

Climbing up the hill



After the first "valley", the numerical bootstrap starts finding solutions where the correlator is infinite, a "projective limit" where only the ratios of OPE coefficients make sense, until the second "valley".

A projective limit

Solutions without the identity operator exist. For example:

$$G(u,v) = (uv)^{-\Delta_\phi/2} + u^{-\Delta_\phi/2} + v^{-\Delta_\phi/2}$$

is manifestly crossing symmetric, and could be unitary at special values of external weight above $\Delta_\phi \geq d-2$.

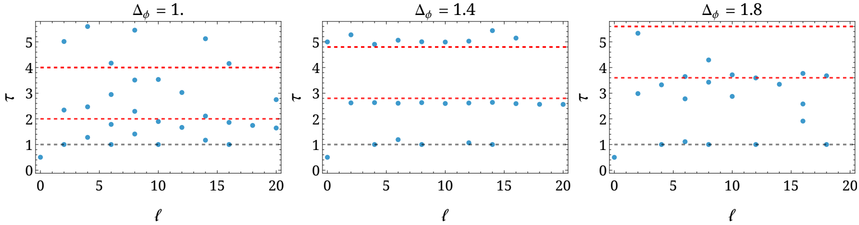
The bootstrap found solutions where $\frac{|\lambda_{\phi\phi\mathcal{O}}|}{|\lambda_{\phi\phi 1}|} \gg 1$.

In fact, while bounding $\langle \Delta \rangle$, SDP only sees the normalized density

$$\tilde{p}_\ell(\Delta) = \frac{\sum_{\mathcal{O}} \lambda^2_{\phi\phi\mathcal{O}}\, g_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z})\delta(\Delta - \Delta_{\mathcal{O}})\delta_{\ell,\ell_{\mathcal{O}}}}{G(u,v)}.$$

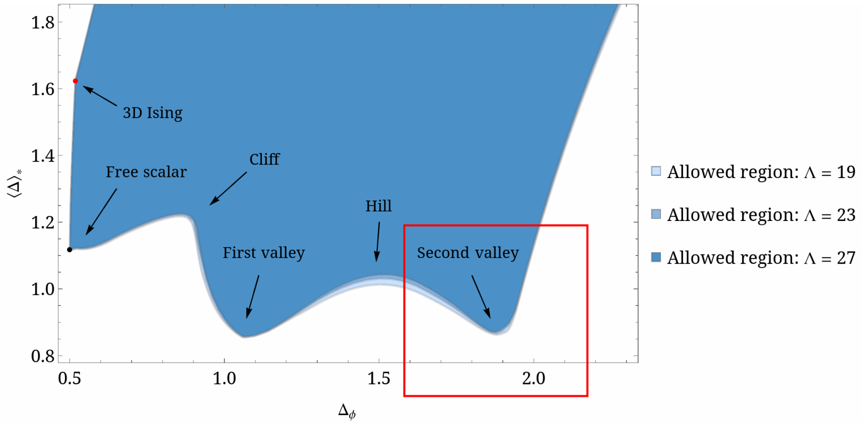
Climbing up the hill

However, the moments computed from the solution above didn't match the lower bound. The "hill" remains a mystery.

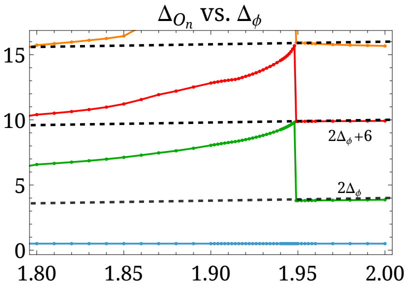


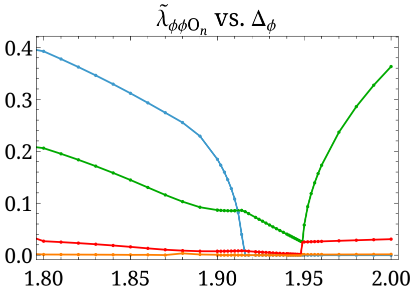
Interestingly, the observed spectral behavior appears to be correlated with the hill-shaped lower bound.

The second "valley"



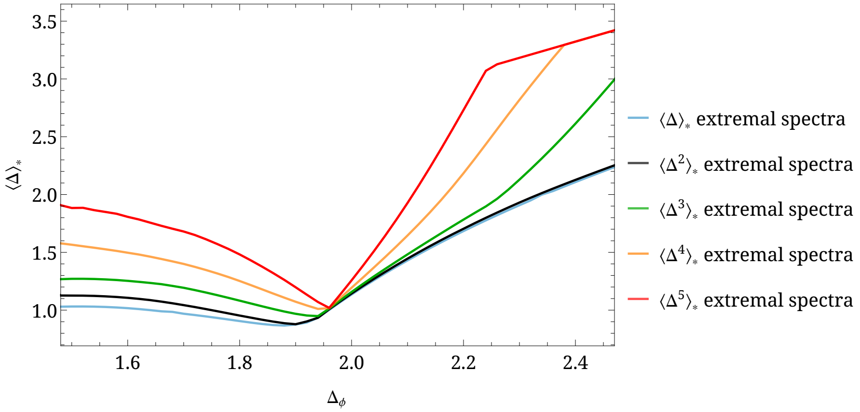
Second valley: the ϕ^2 operator returns





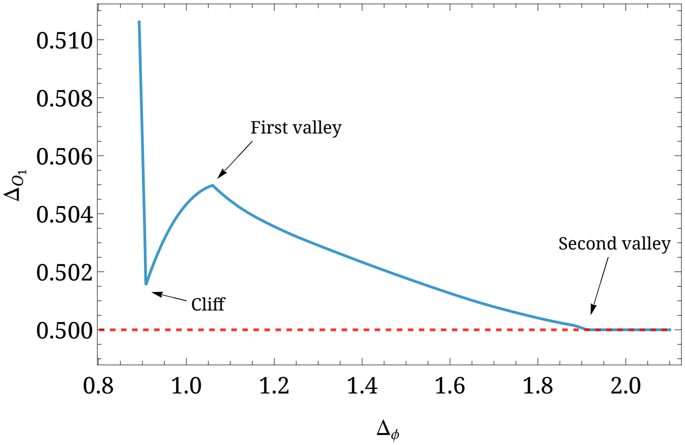
Second valley: simultaneous moment minimization

We use moments as effective probes of the spectral coincidence.



This happens only at this unique point along the lower bound!

Second valley: saturation of unitarity bound



What does it mean? Isn't a scalar at $\frac{d-2}{2}$ a free field that decouples?

The "fake primary" effect

Karateev et al., arXiv:1902.05969

Erramilli et al., arXiv:2011.01959

The residue of a conformal block at its poles is proportional to another conformal block of its null descendant. At the scalar unitarity bound:

$$g_{\Delta,0}(z,\bar{z}) \to \frac{\frac{1}{8d}\left(\frac{d-2}{2}\right)^3}{\Delta - \frac{d-2}{2}} g_{\frac{d+2}{2},0}(z,\bar{z}).$$

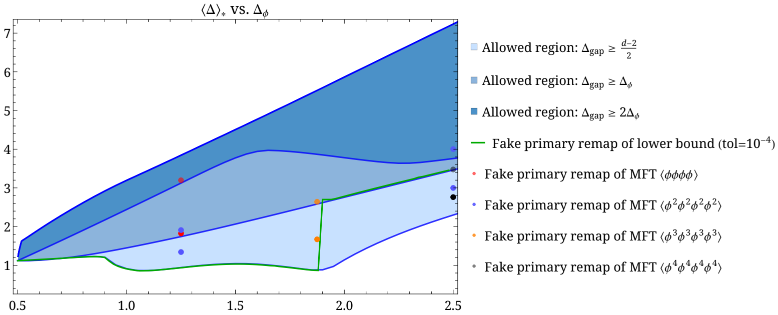
$\Rightarrow$ Subtlety: the product $\lambda^2_{\phi\phi\mathcal{O}} g_{\Delta_{\mathcal{O}},0}$ can remain finite!

$\Rightarrow$ If a theory contains an operator exactly at $\Delta = \frac{d+2}{2}$, it can as well be interpreted as an operator at $\Delta = \frac{d-2}{2}$!

$\Rightarrow$ The correlator and the crossing equation don't see this difference, but the moments do!

A "folded" moment space?

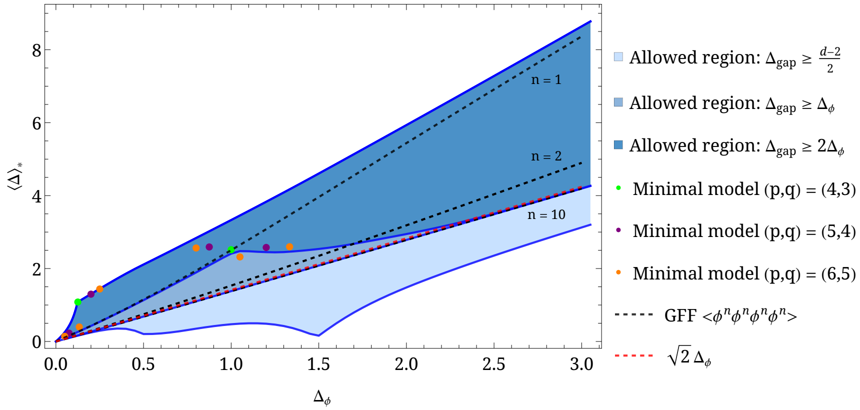
- We move the scalar at $\frac{d-2}{2}$ to its shadow while retaining its contribution to the spectral density.
- The correlator stays the same, but the moment gets lifted.



A point on the boundary is identified with a point in the interior!

An extreme case in $d \to 2^+$

In $d = 2 + 10^{-6}$, the entire lower bound can be lifted.



What are the kink solutions at half-integer external weights?

An extreme case in $d \to 2^+$

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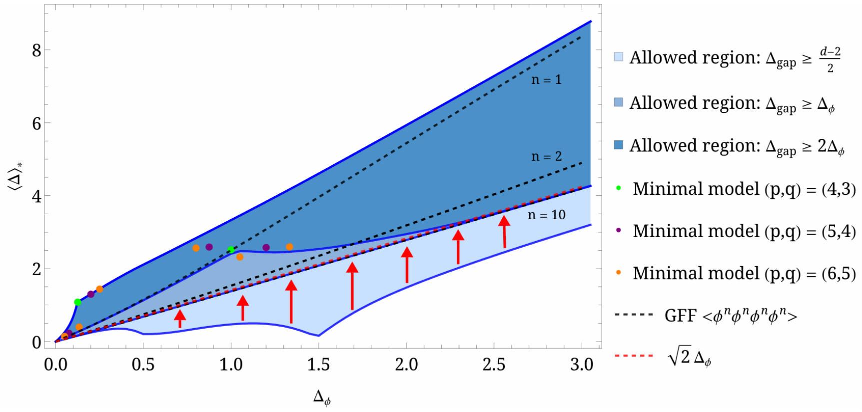
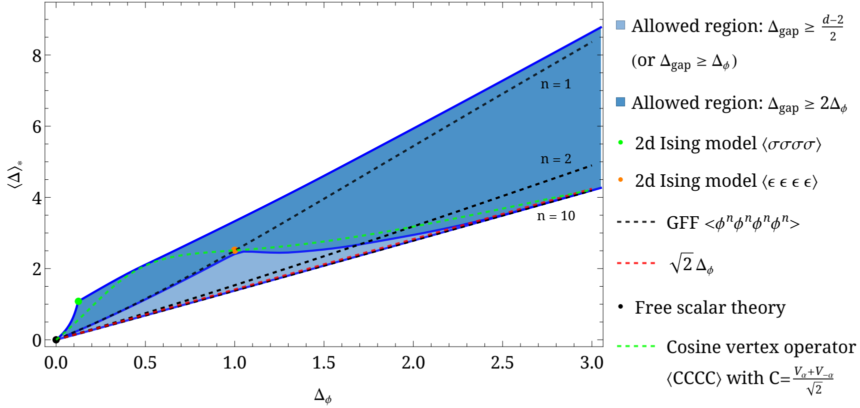


What are the kink solutions at half-integer external weights?

An extreme case in $d \to 2^+$

The lower bound is discontinuous in the limit $d \to 2^+$.



The mechanism stops working here: $g_{\Delta,0} \sim \frac{\frac{1}{8d}\left(\frac{d-2}{2}\right)^3}{\Delta - \frac{d-2}{2}} g_{\frac{d+2}{2},0} \to 0.$

Work in Progress

- Mixed-correlator system might help demystify the kinks.
- From bounding moments to counting operators?
  $\rightarrow$ We can study moments defined on a closed interval.
  $\rightarrow$ A rank certificate for

$$\begin{pmatrix} \nu_0 & \nu_1 & \nu_2 & \\ \nu_1 & \nu_2 & \nu_3 & \cdots \\ \nu_2 & \nu_3 & \nu_4 & \\ & \vdots & & \ddots \end{pmatrix}$$

  would rigorously tell us how many operators must be present!
- Rigorously reconstructing OPE data from moments remains an intriguing open question.
  $\rightarrow$ How can it help us understand the *interior* of the island?
- Generalization to other kinematic configurations?

Conclusions

- Moments provide a coarse-grained but powerful coordinate system for the space of CFTs.
- They recover known heavy-limit and Ising physics while revealing new kinks and spectral reorganizations.
- Future: seek mixed-correlator/analytical understanding of observed structures, and explore applications of moments in precision studies of CFTs.

Thank you!